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RESEARCH ARTICLE

A Novel Hybrid GCN-LSTM Algorithm for Energy Stock Price Prediction: Leveraging Temporal Dynamics and Inter-Stock Relationships

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ABSTRACT Energy stock price prediction is a pivotal challenge in financial forecasting, characterized by high volatility and complexity influenced by geopolitical factors, regulatory shifts, and sector-specific issues. Traditional methods often struggle to account for the intricate dependencies and temporal patterns present in energy stock data. To address these limitations, this study introduces a hybrid model that integrates a Graph Convolutional Network (GCN) with an attention-enhanced Long Short-Term Memory (LSTM) architecture. By employing a graph structure derived from Dynamic Time Warping (DTW), the GCN captures inter-stock relationships, while the attention mechanism within the LSTM component refines the modelling of temporal dynamics, allowing the model to focus on the most relevant historical information. Experimental evaluations across multiple energy stocks show that this combined LSTMGC model significantly outperforms conventional approaches- including Linear Regression, GRU, MLP, and standalone LSTMs- when assessed using Mean Squared Error (MSE) and R-squared (R^2) metrics. By jointly leveraging spatial and temporal dependencies, as well as the selective attention mechanism, the proposed framework enhances predictive accuracy and reliability, offering valuable insights for investors and policymakers navigating the evolving energy market.

INDEX TERMS Stock price, energy, graph neural networks, long short-term memory, dynamic time warping, graph convolutional networks.

I. INTRODUCTION

Predicting energy stock prices has always been a challenging task. The values of oil, gas, and renewable energy companies are influenced by a wide range of factors, including political events, regulatory changes, technological advancements, and market sentiment. These diverse influences make it difficult for traditional financial models to accurately forecast price movements. As a result, researchers and practitioners are increasingly turning to advanced methods- especially those based on Artificial Intelligence (AI) and machine learning (ML)- to better understand and predict trends in this sector.

Recent studies have highlighted the effectiveness of combining both fundamental and technical analysis to improve predictive accuracy. Gurrib et al. found that using a blend of different approaches can outperform models relying on a single method. Meanwhile, the introduction of deep learning techniques has opened new possibilities. Models such as Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs) can identify patterns and relationships in data that traditional models may overlook [1]. As Kim showed, integrating public sentiment into deep learning models can capture psychological factors that affect energy stock prices, helping to refine predictions [2].

Other researchers have explored additional methods and data sources to enhance forecasting. Ma et al. demonstrated that backpropagation neural networks can improve prediction accuracy when applied thoughtfully [3], while Yu and Li emphasized the importance of including a broad range of variables, such as macroeconomic indicators and global

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events [4]. In addition, the use of unstructured information like social media data can offer valuable insights. Mehta et al. showed that sentiment analysis from social media can detect mood shifts in the market, improving a model's ability to foresee price changes [5].

Furthermore, hybrid approaches that combine multiple machine learning techniques and data preprocessing steps have gained popularity. Chen et al., for example, proposed a CNN-BiLSTM-ECA model that leverages both CNNs and LSTMs for improved accuracy [6]. Statistical methods like singular spectrum analysis have also been integrated into neural network models, as demonstrated by Youssef et al., leading to better forecasting performance [7].

In summary, the prediction of energy stock prices now benefits from a wide range of advanced analytical tools. Integrating fundamental and technical analysis, deep learning architectures, sentiment analysis, and hybrid models can help capture the complex and fast-changing nature of these markets. As research in AI and data analytics continues to evolve, there is reason to expect even better forecasting results in the future, ultimately supporting more informed decision-making by investors and stakeholders in the energy sector.

The energy sector, comprising oil, gas and renewable energy sources, holds a critical position in the global economy, exerting substantial influence over financial markets and economic stability. Energy stock prices are inherently volatile, affected by a myriad of factors including geopolitical events, regulatory shifts, technological advancements and changing market demands. Accurately predicting these price movements is crucial for investors, corporations and policymakers, as it supports informed decision-making and effective financial risk management.

Traditional approaches to stock price forecasting, such as linear regression and moving averages, often fail to capture the complex and dynamic nature of financial markets. These methods struggle with the high volatility and intricate interdependencies among energy stocks, leaving a notable gap in handling the complexities of energy markets. In response to these challenges, the integration of DTW with a LSTMGC presents a promising solution. DTW, a time series alignment technique, measures similarity between stock time series by aligning their time points to minimize distance, providing a robust framework for understanding relationships across stocks. Meanwhile, GCNs- a powerful class of machine learning models designed to process graph-structured data- excel at analyzing complex interactions within a network of stocks. This approach is particularly well-suited to modeling the dependencies between various energy stocks and predicting future price movements based on these interconnections. By combining DTW's ability to identify strong inter-stock relationships with GCN's advanced capabilities, we develop the LSTMGC model to address the limitations of traditional methods, aiming to enhance the accuracy and reliability of energy stock price predictions. This integrated solution offers stakeholders a more effective tool for investment planning and risk management in the energy sector.

This study addresses an essential need in financial forecasting by improving predictive accuracy through innovative methodologies. By leveraging DTW and GCN within the LSTMGC model, it seeks to provide deeper insights into the dynamics of energy stocks, supporting more precise investment strategies and fostering resilience in an increasingly interconnected and complex financial environment.

This paper is structured as follows: Section II provides a comprehensive literature review, introducing stock price prediction and discussing traditional methods, machine learning approaches, the role of hybrid GCN-LSTM models in financial forecasting, the application of DTW in time series analysis, and existing hybrid models. The section concludes by identifying gaps in the current literature. Section III describes the proposed hybrid GCN-LSTM model. It outlines the datasets used, data preparation techniques, graph construction for inter-stock relationships, the model's architecture, training processes, and evaluation metrics. Section IV presents the data and experimental results. It evaluates the performance of the proposed model on various stocks and summarizes comparisons with other approaches in terms of accuracy, robustness, and computational efficiency. Section V discusses the results, including deployment considerations and practical implications for the energy stock market. Section VI outlines future research directions, addressing potential improvements, challenges, and unexplored opportunities for enhancing hybrid models in financial forecasting.

II. LITERATURE REVIEW

Stock price prediction remains a challenging and dynamic area of research due to the inherent volatility and non-stationarity of financial markets. Traditional methods, such as Autoregressive Integrated Moving Average (ARIMA), have long been a cornerstone for time series forecasting, offering robust techniques for capturing temporal dependencies in stock price data [1], [2]. However, their limitations in handling non-linear and high-dimensional data have motivated the adoption of advanced machine learning approaches.

Recent advancements in machine learning and deep learning have significantly enhanced stock price prediction accuracy. For instance, Long Short-Term Memory (LSTM) networks, known for their ability to model sequential data and long-term dependencies, have demonstrated superior performance over traditional models, especially in volatile markets [3], [4]. These advancements are further complemented by hybrid approaches that combine multiple techniques to improve predictive performance. One such model integrates Complete Ensemble Empirical Mode Decomposition (CEEMD), LSTM, and LightGBM, outperforming standalone models in both fitting and accuracy [5]. Another notable approach is the incorporation of sentiment analysis from social media platforms, such as Twitter. By analyzing public opinion, models like those combining sentiment analysis with LSTM have successfully captured

market dynamics during periods of high uncertainty [6]. Similarly, optimization techniques have been applied to improve LSTM performance. For instance, the MS-SSA-LSTM model uses swarm intelligence to optimize LSTM hyperparameters, achieving enhanced predictive accuracy [7]. In addition, innovative methods such as the TK-GAN model integrate generative adversarial networks with sentiment analysis. By factoring in public opinion influences, this approach improves the training process and enhances predictive accuracy [8]. Furthermore, utilizing signature path features from limit order book data has shown promising results, particularly in improving prediction efficiency in developed financial markets [9].

Despite these advancements, stock price prediction remains inherently challenging. The Efficient Market Hypothesis (EMH) posits that stock prices reflect all available information, making prediction inherently difficult. However, deviations from EMH provide opportunities for predictive modeling [14]. Addressing these complexities often requires combining advanced methodologies to better capture the non-linear and dynamic nature of financial markets.

A. TRADITIONAL METHODS

The traditional methods of stock price prediction have been foundational in financial analysis, providing frameworks that investors and analysts utilize to forecast future price movements. This literature review synthesizes key approaches, including fundamental analysis, technical analysis, and time series forecasting, highlighting their methodologies, strengths, and limitations.

Fundamental Analysis is a method that evaluates a stock's intrinsic value by examining related economic, financial, and other qualitative and quantitative factors. This analysis often involves scrutinizing financial statements, market conditions, and macroeconomic indicators. For instance, Fitriani et al. emphasize the importance of financial ratios in predicting stock prices, suggesting that a thorough analysis of these ratios can yield insights into future price movements [10]. Similarly, Xiong discusses the role of fundamental analysis in the context of the COVID-19 vaccine industry, illustrating how financial indicators can guide investment decisions [11]. However, the effectiveness of fundamental analysis can be hindered by market efficiency, as noted by Rawlin and Pakalapati, who argue that in efficient markets, accurately forecasting prices based solely on fundamental data can be challenging [12].

Technical Analysis, on the other hand, focuses on historical price movements and trading volumes to identify patterns and trends. This approach utilizes various tools, such as charts and indicators, to predict future price behavior. Ginting and Matondang highlight that technical analysis allows investors to discern trends by statistically analyzing past market data [13]. Moreover, Shiral and Torvi point out that despite the reliance on historical data, technical analysis often struggles with accuracy in predicting stock prices, underscoring its limitations when used in isolation [14]. The use of graphical models, such as Fibonacci retracement, is also noted as a common technique within technical analysis, which attempts to forecast future price movements based on historical price levels [15].

Time Series Forecasting is another traditional method that employs historical data to predict future stock prices. The ARIMA model is one of the most widely used time series models in stock price prediction. It captures the temporal dependencies in stock price data, allowing for effective short-term forecasting. Hsu et al. discuss the significance of ARIMA models in recognizing price patterns, particularly emphasizing the importance of sample size in enhancing prediction performance [16]. However, the limitations of ARIMA models are also acknowledged, particularly in efficient markets where price movements may not follow predictable patterns [17]. The application of ARIMA has been shown to yield varying degrees of success, often depending on market conditions and the specific characteristics of the stock being analyzed [17].

B. MACHINE LEARNING & DEEP LEARNING APPROACHES

The application of machine learning and deep learning (DL) approaches to stock price prediction has gained significant traction in recent years, driven by the increasing complexity of financial markets and the availability of vast amounts of data. This literature review synthesizes key studies that highlight the methodologies, advantages, and challenges associated with these advanced predictive techniques.

1) MACHINE LEARNING APPROACHES

Traditional machine learning methods such as Support Vector Machines (SVM), Random Forest (RF), and Artificial Neural Networks (ANN) have been widely employed for stock price prediction. For instance, Nõu's comparative study on the Baltic stock market demonstrated that SVM and RF models yielded the most accurate predictions among various machine learning algorithms. Similarly, Bai and Sun emphasized the role of sentiment analysis combined with machine learning, noting that social sentiment significantly enhances prediction accuracy in stock markets [18]. The integration of multiple features, including technical indicators and sentiment data, has been shown to improve the robustness of predictions, as evidenced by the findings of Xu and Yang, who applied various ML models to predict Tesla's stock prices [19].

2) DEEP LEARNING TECHNIQUES

Deep learning methods, particularly LSTM networks, have emerged as powerful tools for time series forecasting in stock price prediction. Wang et al. highlighted the effectiveness of LSTM in capturing complex patterns in stock price movements, outperforming traditional models such as

linear regression [20]. In a similar vein, Peng's investigation into China's new energy vehicle sector revealed that LSTM models provided superior predictive performance compared to conventional econometric models [21]. The ability of LSTM networks to learn from sequential data makes them particularly suitable for financial time series, where past prices influence future movements.

C. GRAPH NEURAL NETWORKS IN FINANCIAL FORECASTING

Graph Neural Networks (GNNs) have emerged as a powerful tool in financial forecasting, leveraging their ability to model complex relationships and dependencies inherent in financial data. The application of GNNs in this domain is particularly relevant due to their capacity to capture both spatial and temporal dependencies among various financial entities, such as stocks, commodities, and economic indicators. One of the primary advantages of GNNs is their ability to handle multivariate time series data effectively. Traditional forecasting methods often struggle to fully exploit the interdependencies among multiple variables. Wu et al. highlight that GNNs can model these relational dependencies more effectively than conventional methods, enhancing the accuracy of predictions in fields such as finance and economics [22]. This capability is crucial in financial forecasting, where the performance of one asset can significantly influence others, necessitating a model that can account for these interactions. Furthermore, the robustness of GNNs to topology perturbations is a critical factor in financial applications. Yang et al. discuss how GNNs maintain their predictive performance even when the underlying graph structure is altered, which is particularly relevant in dynamic financial markets where relationships between entities can change rapidly [23]. This robustness allows for more reliable forecasting in environments characterized by volatility and uncertainty. In addition to their structural advantages, GNNs can also incorporate advanced mechanisms such as attention layers, which allow the model to focus on the most relevant features of the data. Xiong's research introduces a complex network attention mechanism that significantly improves the forecasting accuracy of multivariate time series, demonstrating the potential of integrating attention mechanisms within GNN frameworks [24]. This integration can lead to a more nuanced understanding and predictions of financial trends. Moreover, the ability of GNNs to process negative edges—representing adverse relationships—further enhances their applicability in financial forecasting. Ni et al. emphasize that traditional GNNs often fail to capture the complexities introduced by negative edges, which are common in financial networks where certain relationships can be detrimental [25]. Addressing this limitation can lead to more accurate models that reflect the true nature of financial interactions. The scalability of GNNs is another significant advantage, particularly in the context of large financial datasets. Weber's work on scalable graph convolutional neural networks illustrates how these models can efficiently analyze massive and dynamic financial data, making them suitable for real-world applications such as anti-money laundering and fraud detection [26]. This scalability is essential for financial institutions that deal with vast amounts of data daily.

D. DTW IN TIME SERIES ANALYSIS

Dynamic time warping is a widely recognized algorithm in time series analysis, particularly noted for its ability to measure similarity between two temporal sequences that may vary in speed. This characteristic makes DTW particularly advantageous in various applications, including speech recognition, financial forecasting, and environmental monitoring. One of the primary strengths of DTW lies in its flexibility to handle non-linear alignments between time series. Unlike traditional methods such as Principal Component Analysis (PCA) or correlation coefficients, which often assume linear relationships and fixed time intervals, DTW allows for dynamic shifts in time points. This flexibility is crucial in scenarios where time series data exhibit asynchronous behavior or varying lengths, as highlighted by Sakai and Inenaga, who emphasize DTW's capability to compute optimal one-to-many alignments [27]. This property enables DTW to capture complex temporal dynamics that are often overlooked by other methods, thereby enhancing the accuracy of analyses in fields like production forecasting and climate studies [28]. Moreover, DTW's robustness to noise and distortions in time series data further solidifies its utility. For instance, Dilmi et al. demonstrate the effectiveness of DTW in comparing rainfall time series, showcasing its ability to provide a pertinent measure of dissimilarity even in the presence of irregularities [29]. This robustness is particularly beneficial in environmental applications, where data may be affected by various external factors. In the context of time series classification, DTW has proven to be an effective distance measure. Mahajan introduces a novel approach called Graph-Theoretic Dynamic Time Warping (GT-DTW), which enhances traditional DTW by representing time series as graphs. This method not only improves computational efficiency but also provides superior classification accuracy compared to conventional DTW methods [30]. Such advancements are critical as they address the computational challenges associated with DTW's quadratic time complexity, which can hinder its application in large datasets [31]. Furthermore, the integration of DTW with machine learning techniques has opened new avenues for its application. For example, Yu et al. propose a gated recurrent neural network that incorporates DTW for daily peak load forecasting, demonstrating how DTW can enhance predictive modeling in complex, non-linear time series [32]. This synergy between DTW and deep learning illustrates the algorithm's adaptability and relevance in modern analytical frameworks.

E. HYBRID MODELS FOR STOCK PREDICTION

Hybrid models for stock prediction have gained significant attention in recent years due to their ability to combine

various methodologies, enhancing predictive accuracy and robustness. These models often integrate machine learning techniques with traditional statistical methods, deep learning architectures, and sentiment analysis to capture the complex dynamics of financial markets. One prominent approach is the integration of LSTM networks with other machine learning algorithms, such as LightGBM. Ye et al. propose a hybrid model that utilizes LSTM for capturing temporal dependencies in stock price movements while employing LightGBM to process historical stock data, thereby improving prediction accuracy for stock rises and falls [33]. This combination leverages the strengths of both models: LSTM's ability to handle sequential data and LightGBM's efficiency in managing large datasets. Another notable hybrid approach is presented by Ilyas et al., who develop a model that combines Support Vector Regression (SVR) with a modified Hodrick-Prescott filter. Their findings indicate that this hybrid model outperforms traditional neural network methods in forecasting stock closing prices, showcasing the effectiveness of integrating filtering techniques with machine learning [34]. This approach highlights the importance of preprocessing data to enhance the performance of predictive models. In the realm of sentiment analysis, Bai and Sun emphasize the role of social sentiment in stock market predictions. Their intelligent model incorporates sentiment analysis alongside machine learning techniques, demonstrating that social sentiment can significantly improve the accuracy of stock price predictions [18]. This integration of qualitative data with quantitative models represents a crucial advancement in understanding market movements. Moreover, the work of Jiang et al. introduces a hybrid model that combines deep learning with investor sentiment analysis. This model effectively captures the influence of market sentiment on stock prices, providing a more comprehensive prediction framework that accounts for both quantitative and qualitative factors [35]. Such models are particularly relevant in today's market, where investor sentiment can heavily influence stock price movements. The application of hybrid models is not limited to LSTM and SVR; Lu et al. explore the use of convolutional neural networks in conjunction with LSTM for stock price prediction. Their research indicates that the CNN-LSTM model can effectively analyze multivariate time series data, leading to improved forecasting performance [36]. This hybrid approach capitalizes on CNN's strengths in feature extraction and LSTM's capabilities in sequence prediction. Furthermore, Kumar et al. discuss the integration of evolutionary algorithms with traditional econometric models, resulting in a hybrid system that enhances stock price forecasting accuracy. Their findings suggest that combining different methodologies can yield better results than relying on a single approach [37]. This trend of hybridization reflects a broader movement in financial forecasting towards more sophisticated models that can adapt to the complexities of market behavior.

While significant advancements have been made in stock price prediction through traditional, machine learning, and hybrid approaches, several challenges remain. The inherent volatility and non-linearity of stock markets, coupled with the dynamic interdependencies between stocks, present ongoing hurdles for accurate forecasting. The limitations of data quality, model interpretability, and scalability further complicate the development of robust predictive systems. Additionally, integrating complex temporal and relational dynamics, as required for energy stock prediction, demands innovative algorithms that balance computational efficiency and accuracy. Addressing these challenges is essential to advancing the field and unlocking the full potential of predictive models in financial markets.

III. PROPOSED MODEL

This section introduces the proposed model, detailing its graph-based structure, architecture, training process, and evaluation metrics used to assess performance.

A. MODEL STRUCTURE

The proposed model for energy stock price prediction integrates GCN with LSTM networks to effectively tackle the inherent challenges in financial forecasting. This hybrid architecture leverages the strengths of GCNs in capturing spatial dependencies and LSTMs in modeling temporal patterns, thereby offering a comprehensive framework for accurate predictions.

1) GRAPH CONVOLUTIONAL NETWORK COMPONENT

The GCN component constructs a graph-based structure where each node represents an energy stock, and edges signify the relationships between these stocks. These relationships are derived from historical price correlations, with sector-specific interdependencies enhancing the contextual understanding of market dynamics. By modeling these spatial dependencies, the GCN captures how stocks within the same sector or influenced by similar external factors (e.g., crude oil price fluctuations, geopolitical events) interact and influence each other's price movements.

2) LSTM NETWORK COMPONENT

Complementing the GCN, the LSTM network processes sequential data to identify and learn temporal patterns essential for stock price forecasting. Its gated architecture adeptly handles both short-term fluctuations and long-term dependencies, mitigating issues like vanishing gradients. To enhance the model's capability in focusing on the most relevant parts of the data, we introduce an attention mechanism within the LSTM layers. This addition allows the model to dynamically weigh the importance of different time steps, improving its ability to capture significant temporal patterns and anomalies that may influence stock prices. The attention-enhanced LSTM not only boosts predictive accuracy but also provides better interpretability by highlighting which historical data points the model prioritizes during prediction.

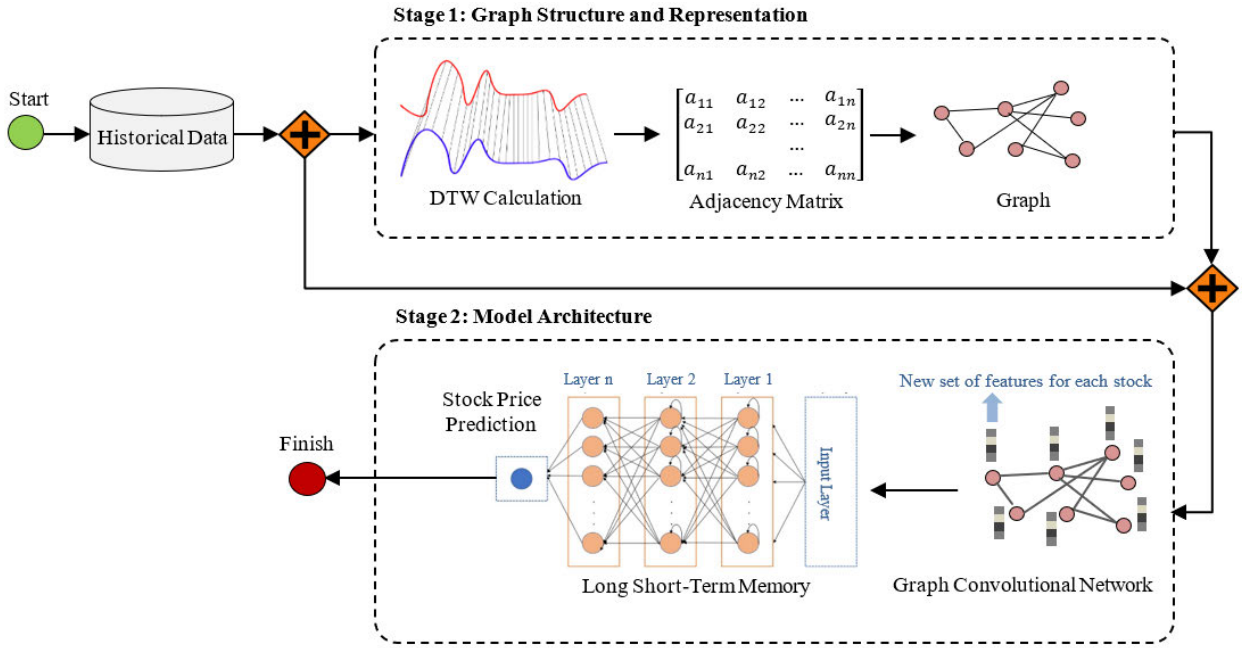


FIGURE 1. Proposed model.

3) DTW FOR ENHANCED CORRELATION MEASUREMENT

Unlike traditional models that treat stocks as isolated entities, our approach employs Dynamic time warping to measure similarities in time series data. DTW effectively aligns time series that may be out of phase, capturing cross-stock dependencies even when stock movements are not perfectly synchronized. This dynamic alignment enriches the graph structure with more meaningful connections, leading to improved predictive performance over models like GRU, MLP, and standalone LSTM, as evidenced by superior metrics such as Mean squared error and R-squared values.

4) OVERALL FRAMEWORK

By combining GCNs with attention-enhanced LSTMs and leveraging DTW for sophisticated correlation analysis, the proposed LSTMGC model offers a robust and versatile tool for forecasting energy stock prices. This integrated approach not only captures the intricate spatial and temporal dynamics of the energy market but also introduces mechanisms for enhanced focus and interpretability, providing valuable insights for investors and policymakers navigating the complexities of the energy sector. Fig.1 illustrates the structure of the proposed model.

Stage 1: Graph Structure and Representation

The relationships between energy stocks are complex and are often influenced by various factors, including market dynamics, geopolitical issues and sector-specific trends. A key challenge in predicting the future prices of energy stocks lies in capturing the interdependencies among these stocks. In this study, we represented energy stocks as a graph, where each stock serves as a node and edges between nodes indicate relationships between stocks. The relationship between any two stocks was quantified by the historical price correlation, providing insights into how these stocks tend to move together over time. This graph-based approach allows us to go beyond analyzing each stock as an isolated time series. By incorporating information about how stocks interact with each other, including temporal alignments that capture shifts and delays in stock movements, we enhance the accuracy of our predictions. This approach reveals underlying connections that may not be immediately apparent, such as coordinated trends or sector-specific dependencies, providing a more nuanced understanding of relationships within the same sector or market. For example, companies in the oil and gas sector often respond similarly to external factors such as changes in crude oil prices, geopolitical events, or shifts in regulatory policies. By representing these interdependencies in the graph structure, the model captures how such factors simultaneously influence multiple stocks, enabling more accurate and context-aware predictions of interconnected stock behavior.

a: CORRELATION MATRIX CONSTRUCTION (USING DTW)

To capture the relationships among energy stocks, we constructed a correlation matrix based on their historical price movements. This matrix serves as the foundation for the graph representation, where each element C_ij represents the correlation between stocks i and j, calculated using DTW. DTW allows us to measure similarities between time series that may not be perfectly aligned in time, making it

particularly valuable for financial data, where stock prices may not move simultaneously but still exhibit similar trends over time. The key advantage of DTW is its ability to handle local variations in speed and duration between compared time series [17]. In Fig.2, a comparison of trends between two different stocks is illustrated.



FIGURE 2. A comparison of trends between PCCYF and XOM stocks.

DTW employs nonlinear mapping to align time series data [38], allowing it to uncover patterns that might not be apparent through simple linear comparisons [39]. This approach is especially useful in identifying underlying similarities among stocks, even when there are time lags or shifts in the price movements, which are common in financial data.

The main idea behind DTW is to stretch or compress portions of a time series to achieve optimal alignment, thereby minimizing the overall distance between the two sequences. This technique calculates a distance metric that accounts for temporal discrepancies by creating an ''alignment path'', which represents the optimal matching between the series.

The DTW calculation begins with the construction of a cumulative distance matrix that compares two sequences $A = (a_1, a_2, \ldots, a_n)$ and $B = (b_1, b, \ldots, b_n)$. The goal is to determine an optimal alignment path $P = (p_1, p, \ldots, p_k)$ where each point $P_k = (i_k, j_k)$ maps elements of A to elements of B, minimizing the overall distance between the sequences. The steps of the DTW algorithm involve constructing a local distance matrix, calculating a cumulative cost matrix and identifying an optimal alignment path that minimizes the overall distance between two time series.

- **Distance Matrix Construction:** First, a local distance matrix $D(i, j)$ is created, where $D(i, j)$ represents the Euclidean distance between two points in the time series:

$$D(i, j) = \sqrt{(a_i - b_i)^2} \tag{1}$$

If the time series lengths are n and m, the distance matrix will have dimensions $n \times m$.

- **Cumulative Cost Matrix Construction:** Next, a cumulative cost matrix $C(i, j)$ is calculated based on the local distance matrix, where the cumulative cost at each point represents the minimum cumulative distance required to align the time series up to that point:

$$\begin{aligned} &C(i, j) \\ &= D(i, j) \\ &\quad + \min\{C(i-1, j), C(i, j-1), C(i-1, j-1)\} \end{aligned} \tag{2}$$

Boundary conditions for $C(1, 1) = D(1, 1)$ are defined and values outside the matrix dimensions are set to infinity. This recursive approach provides a dynamic programming solution that minimizes total alignment cost.

- **Alignment Path:** The optimal alignment is derived from the cumulative distance matrix by tracing back from $C(n, m)$ to $C(1, 1)$. This path is known as the alignment path and its cumulative cost represents the overall similarity between the two-time series. The DTW distance, denoted as $C(n, m)$, reflects the minimum cumulative distance required to align the series.

The Dynamic time warping algorithm effectively compensates for temporal discrepancies between two time series by aligning them in a nonlinear manner. This alignment facilitates the computation of a distance metric that quantifies the similarity between the time series, which we subsequently transform into a correlation value for populating the correlation matrix. In this matrix, stocks exhibiting higher correlation scores are represented by stronger connections within the graph, whereas those with lower scores display weaker or no connections.

The computation of the correlation matrix involves the following steps:

- **Data Input:** We utilize historical price data from the training set encompassing 50 energy stocks.
- **Pairwise Comparison** For each unique pair of stocks i and j, the DTW algorithm is applied to their respective time series to calculate the distance metric D_{ij}.
- **Distance to Correlation Conversion:** The computed DTW distance D_{ij} is inversely transformed into a correlation value C_{ij} using an appropriate scaling function to ensure that smaller distances correspond to higher correlations.
- **Thresholding**: A correlation threshold of 0.7 is implemented, whereby only pairs of stocks with $C_{ij} \geq 0.7$ are retained. This thresholding step ensures that the graph encapsulates only significant relationships, thereby minimizing the inclusion of noise from weaker, potentially irrelevant correlations.

By applying this threshold, the resulting graph structure emphasizes meaningful inter-stock relationships, enhancing the model's ability to capture the underlying market dynamics pertinent to energy stock price prediction.

b: GRAPH ADJACENCY MATRIX AND STORAGE OF GRAPH INFORMATION

After calculating the correlation matrix, we constructed the graph adjacency matrix. This square matrix, denoted as A, has

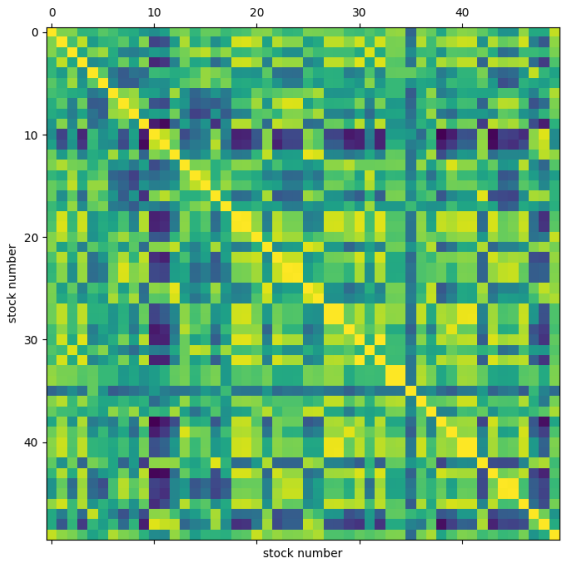


FIGURE 3. Heatmap of relationships between stocks.

elements A_{ij} that indicate whether there is an edge between stocks i and j. If the correlation between two stocks exceeds a threshold of 0.7, we assign a value of 1 to A_{ij}, representing a direct link between these stocks. Otherwise, A_{ij} is set to 0, indicating no direct relationship between the two stocks.

The adjacency matrix serves as a critical input for the graph convolutional layers in our model. It provides information about the graph structure, helping the model determine how to aggregate information from neighboring nodes (stocks) during the convolution operations. This structure enables the model to learn from both individual stocks and their neighbors in the graph, enhancing its predictive power.

To account for varying strengths in the relationships between stocks, we applied a Gaussian kernel to weight the edges in the graph. The Gaussian kernel is defined as follows:

$$A_{ij} = \begin{cases} exp\left(-\frac{d\,(i,j)^2}{\sigma^2}\right) & if\ exp\left(-\frac{d\,(i,j)^2}{\sigma^2}\right) > \varepsilon \\ 0 & otherwise \end{cases} \tag{3}$$

where $d\,(i,j)$ represents the distance between the time series of stocks i and j; σ^2 controls the width of the Gaussian kernel and ε is a threshold that determines whether a meaningful edge exists between the stocks.

The weighted adjacency matrix allowed us to not only model the connections between stocks but also account for the strength of these connections. This weighting played an important role in the graph convolutional layer, where each node collected information from its neighbors based on these weights.

To efficiently manage and utilize the graph structure and its associated attributes, we encapsulated this information within a dedicated class object named GraphInfo. This class was meticulously designed to store and organize essential details, ensuring seamless access and integration during the model training process. The key components stored within GraphInfo include:

- **Node Indices:** Each node index uniquely represents an energy stock within our dataset, facilitating easy identification and reference throughout the model.
- **Edge Indices:** These indices denote pairs of nodes (stocks) that exhibit correlations exceeding the predefined threshold of 0.7. By storing only significant edges, the graph maintains a focused representation of meaningful inter-stock relationships.
- **Total Number of Nodes:** This attribute specifies the total count of stocks included in the graph, which is 50 in our case, providing a clear scope of the dataset being analyzed.

This encapsulated structure ensured that the graph information was readily accessible throughout the model training and could be seamlessly integrated with the graph convolutional layers.

Stage 2: Model Architecture

Our predictive model integrates the strengths of GCNs and LSTM networks, augmented with an attention mechanism to enhance predictive performance. This architecture is specifically designed to capture both spatial dependencies among energy stocks and temporal dependencies within their historical price data.

The first component of the model is a Graph Convolutional Layer, which captures spatial dependencies among energy stocks represented as nodes in a graph. This layer processes the adjacency matrix- constructed from stock correlations- and the stock price data formatted as time series.

For each stock (node), the initial features corresponding to its historical prices are transformed into a higher-dimensional space using a learnable weight matrix. The features of neighboring stocks are then aggregated based on the graph structure, where the influence of each neighbor is weighted by the strength of its connection in the adjacency matrix. This aggregation allows the model to learn the relational structure between stocks, effectively incorporating information from both the individual stock and its neighbors to produce an updated feature representation.

The updated features from the Graph Convolutional Layer are subsequently fed into the LSTM layer to model temporal dependencies. LSTMs are a type of Recurrent Neural Network (RNN) adept at capturing long-term dependencies and patterns in sequential data. The LSTM layer processes the time series data for each stock over a sequence of time steps, updating its hidden state at each step based on the current input and the previous hidden state. This capability allows the LSTM to retain memory of past information, which is crucial for predicting future stock prices based on historical patterns.

To enhance the model's ability to focus on the most relevant temporal features, we incorporate an attention mechanism within the LSTM layer. The attention mechanism enables the model to dynamically assign different weights to different

time steps in the input sequence, effectively highlighting the most influential historical data points for the prediction task.

Specifically, the attention mechanism computes a context vector as a weighted sum of the LSTM outputs, where the weights (attention scores) are learned parameters that indicate the relevance of each time step. This context vector captures the salient features necessary for accurate prediction, allowing the model to focus on significant temporal patterns while diminishing the impact of less relevant information.

By integrating the attention mechanism, the model not only improves predictive accuracy but also enhances interpretability. It provides insights into which historical time steps have the most significant influence on future stock prices, aiding investors and analysts in understanding market dynamics.

The combination of the Graph Convolutional Layer, the LSTM layer, and the attention mechanism allows the model to simultaneously capture spatial dependencies among stocks, temporal dependencies within stock price data, and focus on the most relevant temporal features. This hybrid approach leverages inter-stock relationships and the dynamic temporal patterns of individual stocks, resulting in a more robust and accurate predictive model for energy stock prices.

B. MODEL TRAINING AND OPTIMIZATION

This section focuses on the process of training the predictive model and fine-tuning its parameters to ensure optimal performance. This stage integrates advanced techniques, including the use of GCNs and LSTM layers, requiring careful attention to both spatial and temporal dependencies in the dataset. The following subsections outline the training settings and the hyperparameter tuning process that enabled the development of a robust and generalizable model.

1) TRAINING SETTINGS

The training process was meticulously designed to ensure that the model effectively learned from historical data while maintaining the ability to generalize to unseen data. We utilized the TensorFlow and Keras frameworks to implement and train our model, which integrates GCNs with LSTM layers. This combination required careful tuning of hyperparameters to balance the learning of spatial dependencies among energy stocks (nodes) and the temporal dependencies present in historical price data.

The dataset was divided into three subsets: 70% for training (2301 samples), 20% for validation (657 samples) and 10% for testing (330 samples). This partitioning strategy ensured that the model had ample data to learn intricate stock price patterns from the training set, while the validation set was used for hyperparameter tuning and preventing overfitting. The test set remained untouched during training and validation, serving solely to evaluate the model's final performance and its ability to generalize to new, unseen data.

2) HYPERPARAMETER TUNING

To optimize the model's performance, we conducted extensive experiments with various hyperparameters, including batch size, learning rate, the number of LSTM units, and the type of graph convolution aggregation. The specific steps in our hyperparameter tuning process are as follows:

- **Batch Size:** We tested batch sizes of 32, 64, and 128 to determine the optimal number of samples processed simultaneously during training. This helped in balancing training speed and model performance.
- **Learning Rate:** To prevent the model from overshooting the optimal solution during gradient descent, we explored learning rates of 0.001, 0.0005, and 0.0002. Selecting an appropriate learning rate was crucial for stable and efficient convergence.
- **LSTM Units:** We experimented with the number of LSTM units set to 32, 64, and 128. This was aimed at identifying the capacity needed for effectively capturing temporal patterns in the data.
- **Graph Convolution Aggregation:** The type of graph convolution aggregation was another critical hyperparameter, with options including "mean," "sum," and "max." This parameter influenced how information from neighboring nodes was aggregated, thereby impacting the model's ability to learn the relationships between stocks.
- **Attention Mechanism Settings:**
  - **Number of Attention Heads:** We tested 2, 4, and 8 attention heads to determine the optimal focus granularity.
  - **Attention Dropout Rate:** Dropout rates of 0.1, 0.3, and 0.5 were explored to prevent overfitting within the attention layers.
  - **Attention Vector Dimensionality:** We evaluated attention vector dimensions of 64, 128, and 256 to balance model complexity and performance.
- **Training Epochs and Early Stopping:** We set the number of training epochs to 20, incorporating early stopping based on validation performance to avoid overfitting. Specifically, training was halted if no improvement was observed in the validation loss for 10 consecutive epochs.

Through this rigorous hyperparameter tuning process, the optimal settings were identified as follows:

- Batch Size: 64
- Learning Rate: 0.0002
- LSTM Units: 64
- Graph Convolution Aggregation: "mean"
- Number of Attention Heads: 4
- Attention Dropout Rate: 0.3
- Attention Vector Dimensionality: 128

The model was trained for up to 20 epochs, with early stopping triggered if no improvement in validation performance was detected after 10 epochs. The Mean squared error was employed as the loss function, effectively penalizing larger errors more than smaller ones and encouraging the model to minimize significant deviations between predicted and actual stock prices. For optimization, we selected the RMSProp

optimizer with a learning rate of 0.0002 due to its ability to adapt the learning rate dynamically during training, making it well-suited for handling the evolving patterns in time series data.

The optimization process aimed to minimize the MSE on the training data while ensuring the model's generalizability through validation performance. Early stopping based on validation loss was implemented to prevent the model from overfitting to the training data, ensuring that the final model maintained high performance on unseen data. This comprehensive approach to training and optimization enabled the development of a robust predictive model capable of accurately forecasting energy stock prices by leveraging both spatial and temporal dependencies inherent in the dataset.

C. EVALUATION METRICS

Accurate evaluation of a prediction model is critical to understanding its effectiveness and reliability. In this section, we outline the metrics used to assess the model's performance, focusing on both the magnitude of errors and the model's ability to generalize to unseen data.

1) PERFORMANCE METRICS

To evaluate the performance of the prediction model, we used several metrics commonly employed in time series forecasting. These metrics are as follows:

- **Mean Absolute Error (MAE):** The MAE measures the average magnitude of errors between predicted and actual values, regardless of their direction. It is computed as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{4}$$

  where y_i is the actual stock price and $\hat{y}_i$ is the predicted stock price. The MAE is useful for understanding the average deviation of predictions from actual values in monetary units, providing a straightforward measure of error.
- **Mean squared error:** The MSE calculates the average squared differences between predicted and actual values:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2 \tag{5}$$

  where y_i is the actual stock price and $\hat{y}_i$ is the predicted stock price. Unlike the MAE, the MSE penalizes larger errors more heavily, making it valuable for identifying models that occasionally make large errors.
- **Root Mean Squared Error (RMSE):** The RMSE is the square root of the MSE and expresses the error in the same unit as the predicted variable (stock price):

$$RMSE = \sqrt{\sum_{i=1}^{n}\frac{\left(y_i - \hat{y}_i\right)^2}{n}} \tag{6}$$

  The RMSE is often easier to interpret as it directly relates to the scale of the target variable (stock prices).
- **Coefficient of Determination (R^2):** The R^2 metric evaluates how well the predicted values fit the actual data, indicating the proportion of variance in the target variable explained by the model:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \mu\right)^2} \tag{7}$$

  Here, μ is the mean of the actual values. Higher R^2 values (closer to 1) indicate better model performance.

These metrics allowed us to comprehensively evaluate the model's performance, considering both the magnitude and distribution of the errors.

2) VALIDATION SET EVALUATION

During the training process, a validation set was used to monitor the model's performance at each epoch. The model's error on the validation set was carefully tracked and training was stopped when the validation error ceased to improve. This ensured the model did not overfit the training data and maintained its ability to generalize to unseen data.

At the end of each epoch, the MAE, MSE and RMSE were calculated for both the training and validation sets. By comparing these metrics, we ensured that the model was not excessively optimized for the training set while retaining its generalization capability on the validation set. This approach balanced training and validation errors, ultimately improving the model's robustness and reliability.

IV. DATA AND RESULTS

A. DATA

This section outlines the datasets and data preparation techniques employed to ensure high-quality inputs for the model.

In this study, historical data for the top 50 energy stocks by market capitalization was sourced from Yahoo Finance. These stocks were selected due to their prominent representation of the global energy sector and their considerable influence on broader economic trends (Fig.4). Collecting data from such high-impact stocks is crucial for developing a model that accurately reflects the dynamics of the energy market. Initially, comprehensive time series data for each stock was retrieved. However, during preprocessing, we identified a discrepancy in the time series lengths across stocks, resulting from differences in listing dates and market availability over the years. To ensure uniformity and comparability across all stocks, we confined the dataset to a fixed time period, specifically from March 1, 2011, to September 26, 2024. This period was carefully chosen to encompass a substantial amount of historical data for each stock, thus providing a robust foundation for analysis while minimizing the potential loss of valuable information.

The data collected is on a daily frequency, with the opening price of each trading day selected as the primary variable for analysis. The choice of daily opening prices aligns with

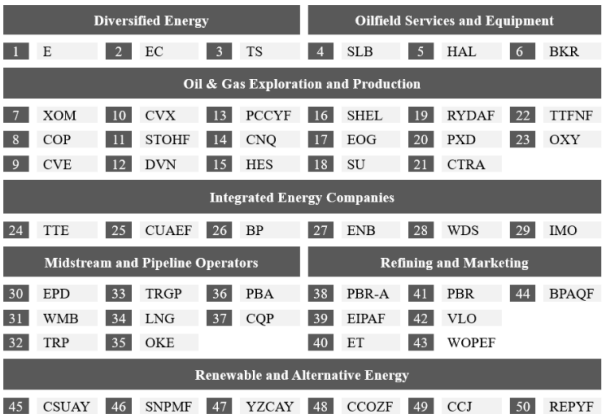


FIGURE 4. The names of the stocks under review, categorized by different energy sectors.

our objective of capturing day-to-day market dynamics and allows for a more granular understanding of price trends and fluctuations over time.

By standardizing the start and end dates for all stocks, we ensured that our model would be trained on a consistent dataset across the entire time series. This alignment not only enhances the model's reliability in capturing sector-wide patterns but also supports accurate comparative analysis across the various stocks in the dataset. Such uniformity is crucial for the validity of the predictive model, as it minimizes biases that could arise from unequal data lengths or irregular time intervals, ultimately contributing to the robustness of our forecasting approach.

B. DATA PREPARATION

Preparing the data is a crucial step in ensuring the accuracy and reliability of stock price prediction models. This involves addressing missing data due to market differences and employing techniques like data smoothing to reduce noise while preserving essential trends in the time series.

Managing missing data is a critical aspect of working with stock prices from global markets, as differences in trading schedules and holidays can lead to inconsistencies in datasets. For instance, some energy stocks in our dataset are traded in non-Western markets, such as China, where holiday schedules often diverge significantly from those of Western markets like the U.S. and Europe. Consequently, missing data points frequently occurred for these stocks on days when their markets were closed but Western markets remained operational. To address these gaps, we adopted the ''forward filling'' method, where missing values are replaced with the last available stock price. This approach ensures the continuity of the time series data while avoiding the introduction of artificial trends or biases that could arise from external interpolations. Forward filling effectively preserves the inherent patterns of the data, making it particularly suitable for financial time series analysis, where maintaining consistency is essential. In the context of our study, forward filling was crucial for aligning and standardizing time series data across 50 energy stocks from diverse markets. This method proved especially reliable given the strong temporal correlations typically observed in stock price movements. By using forward filling, we minimized disruptions caused by trading calendar differences while maintaining the integrity and realism of our dataset.

Moreover, this approach enhances the robustness of predictive models by ensuring they are trained on datasets that closely represent market conditions. The technique aligns with established best practices in time series forecasting and financial analytics, as supported by Kamalov and Sulieman, who highlight its reliability in preserving data fidelity and reducing distortion in sequential datasets [40].

V. RESULTS

To evaluate the performance of the LSTMGC model in predicting energy stock prices, we conducted a comprehensive comparison with several well-established forecasting models. These models included Linear Regression, GRU, LSTM and MLP. The primary goal of this comparison was to assess the ability of each model to predict stock prices by capturing both spatial dependencies (relationships between stocks) and temporal dependencies (time series trends) in the dataset. By comparing the proposed LSTMGC model with these baseline models, we aimed to demonstrate that the integration of Graph Convolutional layers and LSTM layers can lead to significant improvements in prediction accuracy. This comparison highlights the robustness of the LSTMGC model and its ability to generalize to unseen data, which is crucial for real-world financial forecasting applications.

The main objective of this comparison was to determine whether combining Graph Convolutional layers (to capture inter-stock correlations) with LSTM layers (to model time series dependencies) could outperform models that either focus solely on temporal data (e.g., LSTM, GRU) or rely on simpler architectures (e.g., LR, MLP). In the subsequent sections, we will present the performance results of each model, highlighting the strengths of the LSTMGC model. We will also discuss specific areas where baseline models may have demonstrated better performance, providing a balanced evaluation of the proposed approach.

A. RESULTS FOR XOM STOCK

The performance of the LSTMGC model was initially evaluated using the stock prices of Exxon Mobil Corporation (XOM), one of the largest energy companies globally and a key player in the energy sector. The results for XOM were then compared to those of five other models, as shown in Table.1.

The LSTMGC model significantly outperformed all other models across evaluation metrics, as seen in Fig.5. Its Mean squared error was 0.078, indicating the smallest squared differences between predicted and actual prices. Similarly, the Root Mean Squared Error (RMSE) of 0.279 confirms

TABLE 1. Results for different models' prediction applied to XOM stock.

Models	MSE	RMSE	MAE	R^2	MAE %	RMSE %
LSTMGC	**0.078**	**0.279**	**0.224**	**0.815**	**7.383**	**9.203**
GRU	1.899	1.378	1.348	-18.012	85.886	87.829
MLP	2.952	1.718	1.708	-28.558	108.881	109.513
LSTM	1.126	1.061	1.011	-10.275	79.93	82.65
Linear Regression	0.944	0.972	0.933	-8.450	59.475	61.923

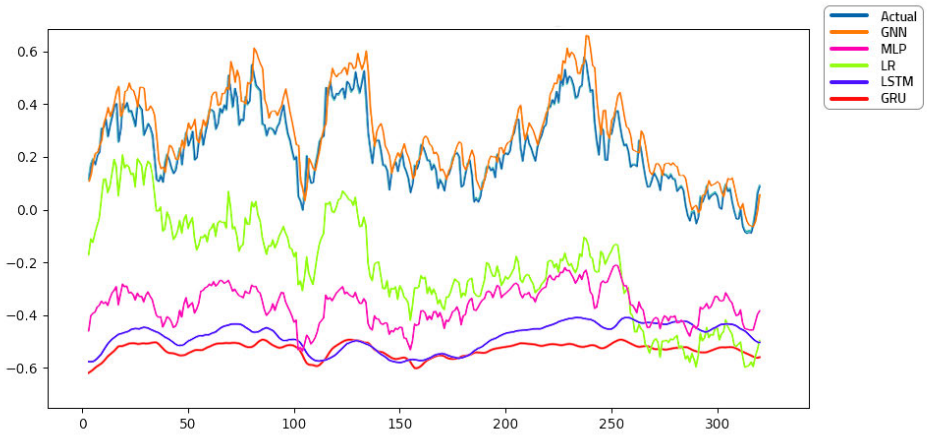


FIGURE 5. The results for different models applied to XOM stock.

the model's high accuracy by demonstrating low average error between predicted and actual prices. Moreover, the mean absolute error of 0.224 suggests that the model's absolute prediction errors were minimal. The high R^2 value of 0.815 highlights that the model explained 81.5% of the variance in the XOM stock prices, showcasing its ability to adapt to the data and make reliable predictions.

The percentage-based error metrics further affirm the model's precision, with the mean absolute percentage error at 7.383% and RMSE% at 9.203%. These metrics indicate that the predicted stock prices deviated only slightly from the actual values. The superior performance of the LSTMGC model can be attributed to its dual capabilities: temporal dependency modeling through LSTM and spatial relationship modeling via GCN, effectively capturing both time-series patterns and inter-stock relationships.

In contrast, the GRU model, while generally effective in time series prediction, performed poorly in this case. Its MSE of 1.899 and RMSE of 1.378 were significantly higher than those of LSTMGC, reflecting substantial prediction errors. With an MAE of 1.348 and an R^2 of -18.012, the GRU model not only exhibited large errors but also failed to capture meaningful variance in the stock prices. The negative R^2 value suggests that the model performed worse than simply predicting the mean stock price, a strong indicator of poor performance. Similarly, its MAPE (85.886%) and RMSE% (87.829%) reveal that the predictions deviated drastically from actual prices. The inability of GRU to account for inter-stock relationships likely contributed to this underperformance.

The MLP model, a feed-forward neural network, fared even worse than GRU. Its MSE of 2.952 and RMSE of 1.718 were the highest among all models, indicating the most significant prediction errors. The MAE of 1.708 further confirms the model's struggle to produce accurate forecasts. With an R^2 of -28.558, the MLP demonstrated very limited capability to explain stock price movements, as expected due to its lack of inherent temporal or spatial dependency modeling. Its MAPE (108.881%) and RMSE% (109.513%) further underscore the model's unreliability, as its predictions deviated massively from actual prices.

The LSTM model, while more effective than GRU and MLP, still lagged behind the LSTMGC model. It achieved an MSE of 1.126 and an RMSE of 1.061, demonstrating some ability to capture patterns in the XOM stock price data but still producing larger errors compared to LSTMGC. Its MAE of 1.011 highlights the relatively larger average prediction deviations. An R^2 of -10.275 indicates that the model struggled to account for the variance in stock prices effectively. Although LSTM is well-suited for capturing temporal dependencies, its inability to model inter-stock relationships likely led to its inferior performance. Furthermore, its MAPE (79.930%) and RMSE% (82.650%) reveal considerable inaccuracy in predictions.

TABLE 2. Results for different models' prediction applied to CVX stock.

Models	MSE	RMSE	MAE	R^2	MAE %	RMSE %
LSTMGC	**0.008**	**0.090**	**0.070**	**0.924**	**-27.039**	**-34.909**
GRU	0.190	0.436	0.372	-1.247	138.077	161.643
MLP	0.121	0.348	0.271	-0.437	100.590	129.265
LSTM	0.127	0.357	0.296	-0.509	96.09	112.81
Linear Regression	0.331	0.575	0.475	-2.916	176.080	213.412

The LR model, while simple, performed better than GRU and MLP but fell short of both LSTM and LSTMGC. Its MSE of 0.944 and RMSE of 0.972 reflect relatively higher prediction errors than LSTMGC. The MAE of 0.933 suggests that while it provided better predictions than GRU and MLP, it was less effective than the more advanced models. The R^2 of -8.450 indicates limited success in explaining stock price variance, as linear models struggle with the non-linear relationships often present in stock price movements. Percentage errors such as MAPE (59.475%) and RMSE% (61.923%) further confirm its reduced accuracy compared to LSTMGC.

Figure 5 clearly illustrates the superiority of the LSTMGC model, which combines temporal and spatial modeling capabilities to provide highly accurate predictions. Other models, particularly GRU, MLP and LR, showed notable limitations in capturing the complex patterns in XOM stock prices, emphasizing the importance of integrating both temporal and spatial features for accurate stock price forecasting.

B. RESULTS FOR CVX STOCK

As shown in Table.2, the LSTMGC model once again outperformed all other models in predicting the stock prices of Chevron Corporation (CVX). With a Mean Squared Error of 0.008 and a Root Mean Squared Error of 0.090, the LSTMGC achieved the lowest error rates, demonstrating its exceptional capability for accurate stock price forecasting. The model also exhibited a Mean Absolute Error of 0.070 and an R^2 value of 0.924, indicating that it explained 92.4% of the variance in CVX stock prices. These results highlight the reliability and precision of the predictions made by the LSTMGC model. Despite some negative percentage-based error values (MAPE: -27.039% and RMSE%: -34.909%), these figures still reflect the model's overall low error levels.

In contrast, other models, including GRU, MLP and LSTM, demonstrated inferior performance. The GRU model, for example, had an MSE of 0.190 and an RMSE of 0.436, which were significantly higher than those of the LSTMGC model, indicating larger prediction errors. Similarly, the MLP and LSTM models produced MSEs of 0.121 and 0.127, respectively, alongside negative R^2 values, further emphasizing their weaker performance. The Linear Regression (LR) model exhibited the poorest performance, with an MSE of 0.331 and an R^2 value of -2.916, underscoring its inability to fit the data effectively.

Overall, the LSTMGC model consistently outperformed the alternative approaches, confirming its ability to leverage both temporal and spatial dependencies in energy stocks like CVX for more accurate predictions. These findings reinforce the importance of combining these features to achieve superior stock price forecasting (Fig.6).

C. RESULTS FOR PCCYF STOCK

As shown in Table.4, the LSTMGC model once again demonstrated superior performance compared to other models in predicting the stock prices of Panasonic Corporation (PCCYF). With a MSE of 0.003 and a RMSE of 0.057, this model achieved the lowest error rates, showcasing its ability to accurately forecast PCCYF stock prices. The MAE of 0.045 and the R^2 value of 0.946 further emphasize its effectiveness, as it explained 94.6% of the variance in PCCYF stock prices, establishing it as a highly reliable tool for prediction. While the percentage-based error metrics were positive (MAE%: 5.887% and RMSE%: 7.422%), these small values confirm the model's accuracy.

In contrast, alternative models such as GRU, MLP and LSTM exhibited significantly higher error rates. The GRU model, for instance, had an MSE of 0.129 and an RMSE of 0.359, reflecting its lower effectiveness in identifying patterns in PCCYF stock prices. The MLP and LSTM models also performed poorly, with MSE values of 0.381 and 0.311, respectively, along with negative R^2 values, indicating their inability to fit the data adequately. While the LR model outperformed GRU, MLP and LSTM, it still displayed relatively high error levels, with an MSE of 0.051 and an R^2 value of 0.242, highlighting its limitations in capturing the stock price movements with precision.

Overall, the LSTMGC model consistently outperformed all other approaches, reaffirming its superior ability to capture both temporal and spatial dependencies in predicting PCCYF stock prices. These results further validate its potential as a robust and reliable forecasting method.

D. SUMMARY OF MODEL COMPARISONS

In the evaluation of all stocks, the LSTMGC model consistently outperformed other models (GRU, MLP, LSTM and

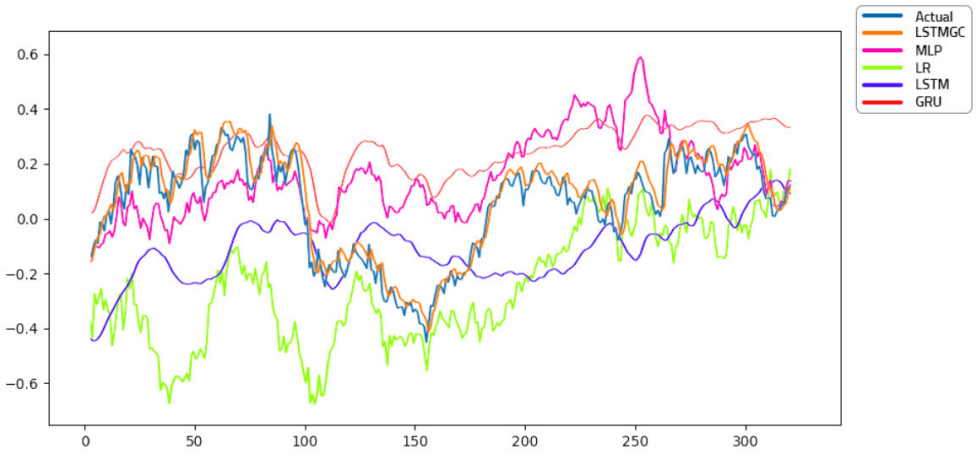


FIGURE 6. The results for different models applied to CVX stock.

TABLE 3. Results for different models' prediction applied to PCCYF stock.

Models	MSE	RMSE	MAE	R^2	MAE %	RMSE %
LSTMGC	**0.003**	**0.057**	**0.045**	**0.946**	**5.887**	**7.422**
GRU	0.129	0.359	0.317	-0.894	38.707	43.828
MLP	0.381	0.617	0.601	-4.601	73.420	75.367
LSTM	0.311	0.557	0.485	-3.570	26.29	32.49
Linear Regression	0.051	0.227	0.199	0.242	24.348	27.719

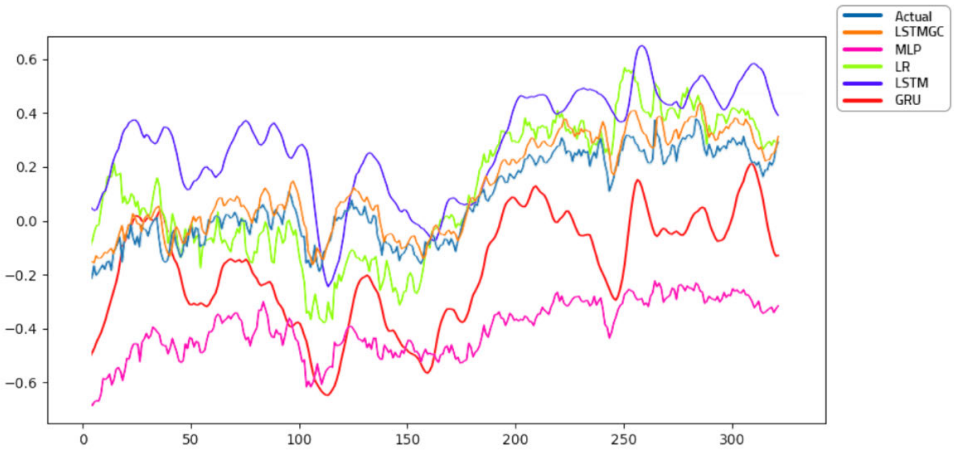


FIGURE 7. The results for different models applied to PCCYF stock.

LR) in terms of prediction accuracy. This superior performance was evidenced by significantly lower error metrics, including MSE, RMSE and MAE. For most stocks, the LSTMGC model achieved the lowest values for these metrics, indicating fewer prediction errors compared to other models. Additionally, the high R^2 values obtained by this model demonstrated its ability to explain a substantial portion of the variance in stock prices. In contrast, the GRU, MLP and LSTM models were unable to match the performance of the LSTMGC model, exhibiting significantly higher error rates and even negative R^2 values, particularly for stocks with more complex price movements. Models such as MLP and Linear Regression, which do not account for temporal dependencies as effectively as recurrent models, consistently demonstrated higher error percentages and lower predictive capabilities.

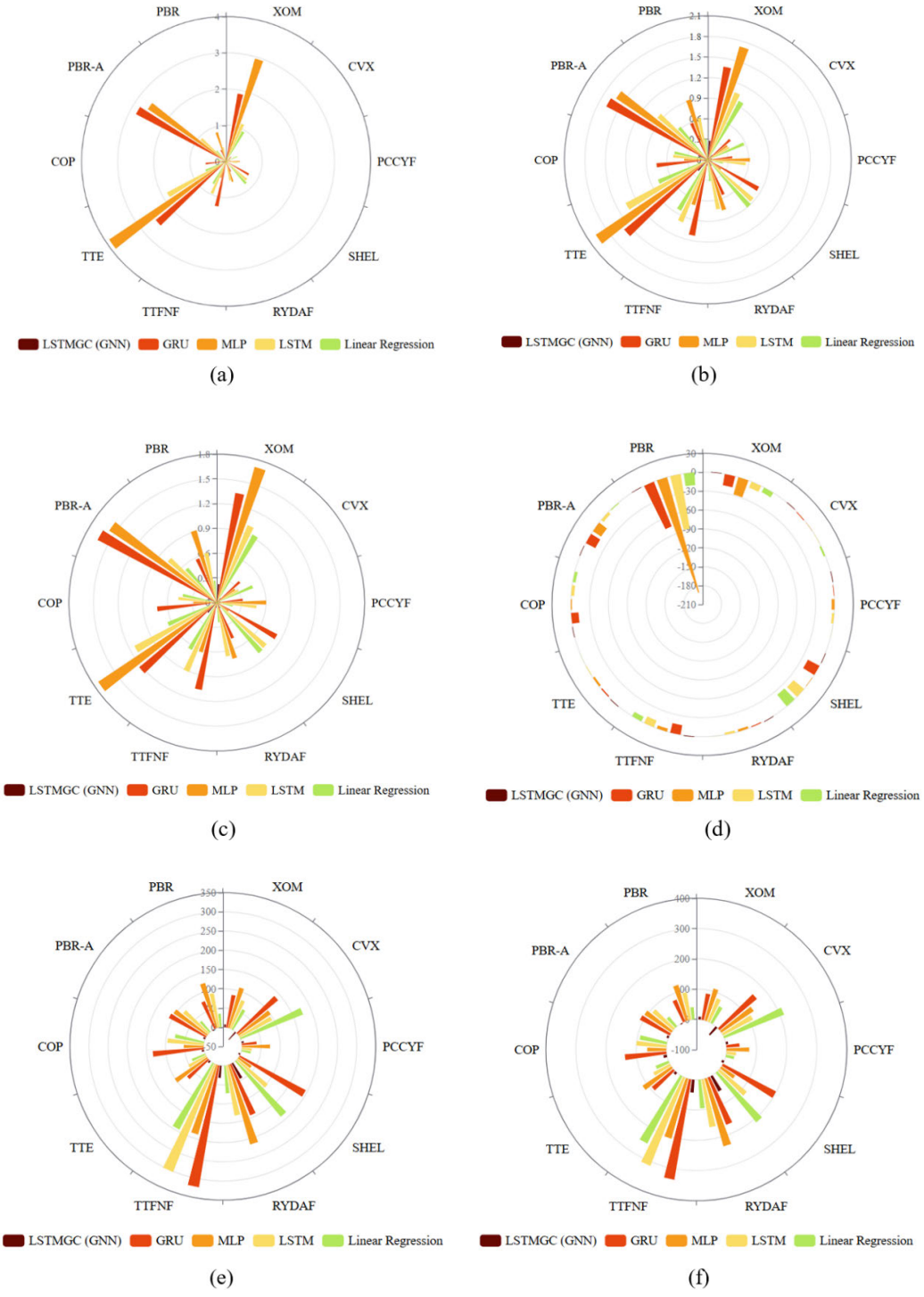


FIGURE 8. (a) Differences in MSE values for each of the nine selected stocks across the five implemented models. (b) Differences in RMSE values for each of the nine selected stocks across the five implemented models. (c) Differences in MAE values for each of the nine selected stocks across the five implemented models. (d) Differences in R^2 values for each of the nine selected stocks across the five implemented models.

The results highlight that the LSTMGC model, which integrates both spatial (inter-stock relationships) and temporal (time series) data, is significantly more robust and effective in predicting energy stock prices. The spatial relationships captured through graph convolution layers provide a distinct advantage over models that rely solely on time-series data



FIGURE 9. Results for different models applied to six other stock: (a) SHEL, (b) RYDAF, (c) TTFNF, (d) TTE, (e) COP, (f) PBR-A, (g) PBR.

or linear relationships. To further illustrate the comparative performance, charts depicting the results of different models for a sample of 10 stocks out of 50 are presented in the following sections.

In the previous sections, a detailed analysis of the forecasted prices for each of the three selected stocks using the five mentioned models was presented, along with corresponding comparative charts. To further evaluate the predictive performance of the proposed model for the ten stocks analyzed (as shown in Fig.8), a comparison chart illustrating the model results for six additional stocks is provided below (Fig.9).Conclusion

The primary goal of this study was to develop and evaluate a robust predictive model for forecasting energy stock prices using a hybrid approach that combines GCN with LSTM networks. Energy stock prices are influenced by numerous factors, such as geopolitical events, commodity price volatility and sector-specific developments, making accurate forecasting a significant challenge. Our approach aimed to address this complexity by leveraging graph convolution layers to capture inter-stock relationships and LSTM layers to model temporal dependencies in stock price data.

A comprehensive comparison with other models, including LR, GRU, MLP and standalone LSTM, demonstrated that the LSTMGC model consistently outperformed the alternatives in terms of prediction accuracy. Specifically, the LSTMGC model exhibited lower error rates and higher R^2 values,

indicating superior capability in explaining the variance in stock price data. The key advantage of the LSTMGC model lies in its ability to capture interdependencies among various stocks. Unlike traditional models that treat each stock as an isolated time series, our approach utilized DTW to represent inter-stock relationships in the form of a correlation matrix. This enabled the model to deliver more precise predictions by incorporating spatial and temporal information.

Despite these promising results, the study revealed certain limitations, particularly in highly volatile market conditions or during sudden geopolitical shocks. While the LSTMGC model performed well across most scenarios, further improvements are required to enhance its robustness and predictive power under such extreme conditions. This research underscores the potential of hybrid deep learning approaches, like LSTMGC, in financial forecasting and highlights future opportunities for refining and expanding their application in dynamic and complex markets.

VI. FUTURE DIRECTIONS

While this study successfully demonstrated the effectiveness of combining GCN and LSTM for energy stock price prediction, there remain several promising avenues for future exploration and enhancement to improve prediction accuracy and broaden the applicability of the model. One significant limitation of the current approach is its reliance solely on historical stock price data. Stock prices are also influenced by external factors, including market sentiment, news, and social media trends. Incorporating sentiment analysis derived from news articles, financial reports, and social media platforms could provide valuable insights into the impact of public perception on stock price movements. By integrating such alternative data sources, especially during periods of market uncertainty, the model's predictive power could be significantly enhanced. Another area of potential improvement lies in generalizing the model to different portfolios beyond the energy sector. This study focused specifically on energy stocks, but applying the LSTMGC model to other sectors such as technology, healthcare, or consumer goods would test its adaptability and generalizability. Investigating cross-sector dependencies and sector-specific dynamics could provide a more comprehensive evaluation of the model's versatility and validate its performance across diverse market conditions. Further, enriching the graph structure used in this study with external variables offers a promising direction for future research. Factors such as commodity prices, macroeconomic indicators, or sector-specific metrics could be incorporated into the graph correlation matrix to better capture complex inter-stock dependencies. A more comprehensive representation of these relationships would likely improve the model's ability to reflect the multifaceted influences on stock prices. Hybrid modeling approaches also present an intriguing opportunity to enhance the robustness and accuracy of predictions. Combining the LSTMGC framework with other predictive techniques, such as Random Forest or XGBoost, in an ensemble model could leverage the strengths of multiple methods. Such a hybrid approach could mitigate the limitations inherent in individual models, improving overall reliability and performance under various market conditions.

Finally, adjusting the time horizons and data granularity used for model training and prediction could yield valuable insights. Exploring higher-resolution data, such as hourly or minute-level data, alongside varying prediction horizons, would allow the model to cater to the needs of both short-term and long-term investors. Understanding how these parameters influence prediction accuracy could refine the model further for practical applications and provide tailored solutions for diverse investment strategies.

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